

EXAM 7 – SPRING 2013

11. (3.5 points)

Given the following information for an insurance company (\$000,000):

	<u>Calendar Year</u>		
	<u>2013</u>	<u>2014</u>	<u>2015</u>
Beginning US GAAP equity	100		
Net income	15	16	17
Minimum capital needed at year-end to maintain current rating	110	120	130

- Risk-free rate: 2.0%
- Expected market return for peer companies: 10.0%
- Insurance company's equity beta: 1.25
- CAPM is an appropriate model for determining the company's risk-adjusted discount rate.
- The company plans to maintain its current rating by meeting the minimum capital requirements.

a. (3 points)

Determine the value of the insurance company as of January 1, 2013 based on the free cash flow to equity method.

b. (0.5 point)

Explain why the free cash flow to equity method is preferable to the free cash flow to the firm method for valuing a property and casualty insurer.

CONTINUED ON NEXT PAGE

Question 11 Sample Answer

Solution 1

a) $k = r_f + \beta (E(r_m) - r_f) = 2\% + 1.25 \times (10\% - 2\%) = 12\%$

$$FCF_1 = 100 + 15 - 110 = 5$$

$$FCF_2 = 110 + 16 - 120 = 6$$

$$FCF_3 = 120 + 17 - 130 = 7$$

$$\text{Assume } g = \text{Capital}_{2015} / \text{Capital}_{2014} - 1 = 130/120 - 1 = 8.3\%$$

$$\begin{aligned} \text{Firm Value} &= \sum FCF / (1+k)^i = 5/(1+12\%) + 6/(1+12\%)^2 + 7/(1+12\%)^3 \\ &\quad + [7(1+8.3\%)/(12\% - 8.3\%)]/(1+12\%)^3 \end{aligned}$$

$$= 160.07 (\$000,000)$$

- b) 1. Debt vs Policyholder Liability: FCFF method firstly calculates total firm value and then subtract debt. There is no economic rationale for different treatments of two kinds of liability.
2. WACC and unlevered return k: FCFF use either WACC or unlevered return for discount rate k. Due to policyholder liability, it is hard to precisely define and estimate either one.

Solution 2

- a) $FCFE = NI + \text{Non-cash charges (0)} + \text{net borrowing (0)} - \text{net working cap (0)} - \Delta \text{ capital req.}$

	2013	2014	2015
FCFE	$15 - (110 - 100) = 5$	6	7

$$k = r_f + \beta (E(r_m) - r_f) = .02 + 1.25 \times (0.01 - 0.02) = 12\%$$

$$\text{need } g = \text{ROE} \times \text{Reinvestment Rate}$$

	2013	2014	2015	selected
$\text{ROE} = \frac{NI}{\text{Beg Eq.}}$	0.15	0.14545	0.14167	0.14167
$\text{Reinv Rate} = \frac{NI - FCFE}{NI}$	0.667	0.625	0.58823	0.58823

Since ROE & Reinv rate are decreasing over years, I will select latest year ROE & Rein. Rate to use to calculate g.

$$\rightarrow g = 0.14167 \times 0.58823 = 0.0833 = 8.33\%$$

$$V_0 = \frac{5}{(1.12)} + \frac{6}{(1.12)^2} + \frac{7}{(1.12)^3} + \frac{1}{(1.12)^3} \times \frac{[7(1.0833)]}{.12-.0833}$$

$$= 14.2299 + \frac{1}{(1.12)^3} \text{TV}$$

$$= 14.2299 + 147.07 = 161.3$$

- b) Because policyholder liability is considered debt, & the FCFF method must incorporate debt into formula. However, incorporating policyholder liability both explicitly & in the WACC is not easy task, and therefore prefer FCFE method.

Solution 3

- a) (1) Determine K with CAPM

$$K = 0.02 + 1.25 \times (0.10 - 0.02) = 0.12$$

(2)

<u>Year</u>	<u>2013</u>	<u>2014</u>	<u>2015</u>
Net Income	15	16	17
Req. Cap.	110	120	130
Increase in Req Cap.	10	10	10
FCFE	5	6	7
Reinvested Cap.	10	10	10

$$\rightarrow \text{FCFE} = \text{NI} - \text{Increase in Req. Capital}$$

For g, we use $\text{ROE}_{2015} \times \text{Reinvestment Rate}_{2015}$

$$= \left(\frac{17}{120}\right) \times \left(\frac{10}{17}\right) = 0.083$$

$$(3) \text{ Terminal Value at 12/31/2015} = \frac{7 \times (1.083)}{0.12 - 0.083} = 204.89$$

$$(4) V = \frac{5}{1.12} + \frac{6}{1.12^2} + \frac{7}{1.12^3} + \frac{204.89}{1.12^3} = 160.07$$

- b) Since policyholder's liabilities account for most of an insurer's debt, it is hard to precisely define the WACC or the discount rate for the APV.

Solution 4

a) $k = r_f + \beta (E(r_m) - r_f)$
 $= 2\% + 1.25 \times (10\% - 2\%) = 12\%$

	2013	2014	2015	
$\Delta \text{ Equity}$	$110 - 100 = 10$	$120 - 110 = 10$	10	
$\text{NI} - \Delta \text{ Equity}$	$15 - 10 = 5$	6	7	
$\text{ROE} = \frac{\text{NI}}{\text{Beg Cap}}$	$\frac{15}{100} = .15$	0.145	0.142	<u>Selected</u> 0.146
ρ	$\frac{10}{15}$	$\frac{10}{16}$	$\frac{10}{17}$	.627

$$\frac{5}{(1.12)} + \frac{6}{(1.12)^2} + \frac{7}{(1.12)^3} = 14.23$$

$$\text{growth} = (.146 \times .627) = .0915$$

$$\frac{7 (1.0915)}{1.12^3 (.12 - .0915)} = 190.82$$

$$190.82 + 14.23 = 205.05$$

- b) FCFE calculates the value of Firm then subtracts the value of debt to the equity value. However, the distinction between policyholder liabilities + debt is arbitrary + there is no economic rationale for treating them differently. The FCFE met alleviates this problem and calculates the equity directly

Solution 5

a) $k = .02 + 1.25 (.1 - .02)$
 $= .12$

	Beginning Equity	FCFE	ROE	Δ FCFE
2013	100	5	.15	
2014	110	6	.145	20%
2015	120	7	.1416	16.667%
Selected			.14	.10

$= NI - \Delta Cap$
 $= 15 - (110 - 100)$

$= \frac{NI}{Beg\ eq.}$

Select a growth rate of 10%

 This seems a little bit high even w/ the low selection

PV Time Horizon

$$= \frac{5}{1.12} + \frac{6}{1.12^2} + \frac{7}{1.12^3} = 14.23 M$$

$$\begin{aligned}
 \text{PV Extended Horizon} &= \frac{7 (1.10)}{.12 - .10} = \frac{FCFE (1+g)}{k-g (1.12)^3} \\
 &= \frac{385}{1.12^3} = 274 \\
 &= \boxed{288.26 M}
 \end{aligned}$$

Examiner Comment

Part a) was generally answered well by candidates.

We accepted several valid methods for calculating the growth rate.

- $g = \text{Reinvestment Rate} \times \text{ROE}$
- $g = \text{FCFE}_t / \text{FCFE}_{t-1} - 1$
- $g = \text{Capital}_t / \text{Capital}_{t-1} - 1 = \Delta \text{Capital} / \text{Capital}_{t-1}$

Full credit was not given if candidates merely assumed a growth rate without calculating and reviewing one of the above metrics.

Some candidates neglected to discount the terminal value, or made other calculation/arithmetic errors.

Some candidates included the beginning capital value in the calculation of the total value of the company. Full credit was not given in these instances.

Many candidates supplied partial answers to part b). Common responses that received partial credit include:

- Mention of taking into account policyholder liabilities for the FCFE method without mentioning that the distinction between debt and policyholder liabilities is arbitrary.
- Mention that the interest rate is hard to define, without explanation (i.e., due to presence of policyholder liabilities)

Candidates that supplied 2 incomplete answers for part b) only received partial credit.